1. After separation of the DNA strands, one strand of DNA, the template strand, acts as a template for RNA polymerase.
2. The enzyme RNA polymerase reads this template strand one base at a time and synthesizes an RNA molecule out of complementary nucleotides always in 5' to 3' direction.
3. The RNA transcript carries the same information as the non-template (coding) strand of DNA the base thymine (T) is replaced by uracil (U).

6.3.2.3 Termination of transcription

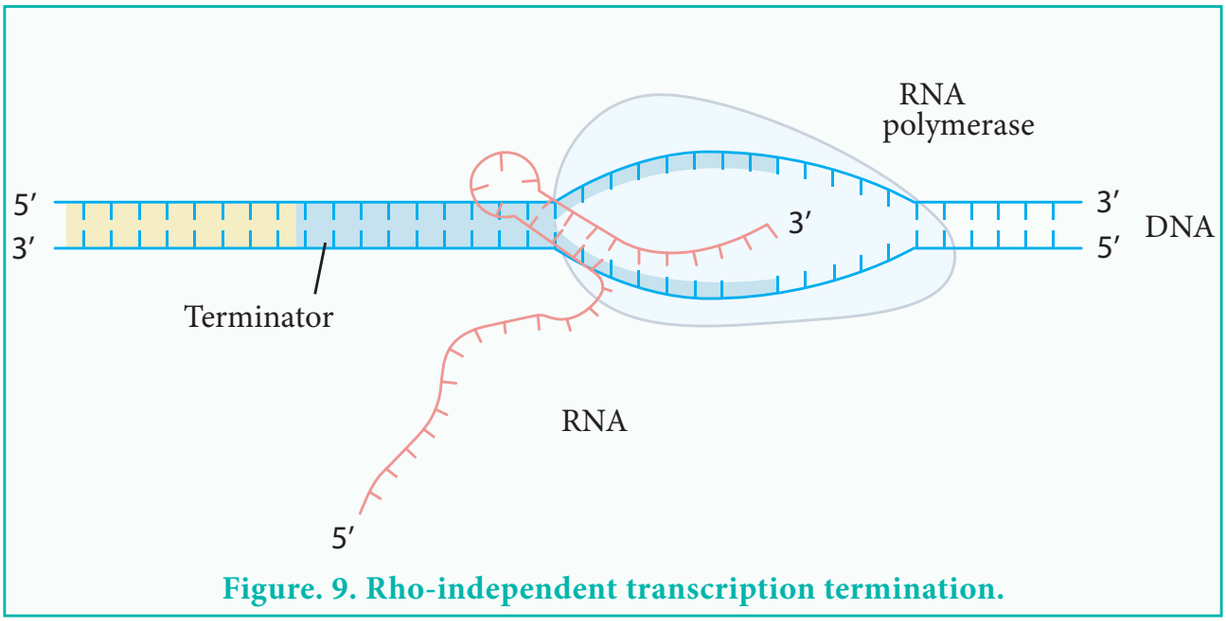

Figure. 9. Rho-independent transcription termination.

1. Two different types of transcription termination are observed in bacteria: the Rho-independent termination and Rho-dependent termination.
2. In Rho-independent termination, the RNA transcript forms a hairpin loop structure rich in GC base pairs, followed by a poly U region.
3. The hair pin structure of the newly synthesized RNA transcript produces a mechanical stress in the RNA polymerase and stalls it from moving. At the same time the relatively weak hydrogen bonds made by the poly U region with template DNA pulls out of the RNA polymerase, terminating transcription.
4. In the "Rho-dependent" type of termination, a protein called Rho destabilizes the interaction between the template and the mRNA and releases the newly synthesized mRNA from the elongation complex.

6.3.3 Types of RNA, synthesis and processing

There are three types of RNA which play a major role in protein synthesis, which are the ribosomal RNA or rRNA, the transfer RNA ot tRNA and the messenger RNA or mRNA

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